



Chapter 2

Systems of Linear Equations

MAE101 - ALGEBRA

Ly Anh Duong

duongla3@fe.edu.vn





Table of Contents

1 Definition

- ▶ Definition
- ▶ Solutions of Systems of linear equations
- ▶ Solving Systems of linear equations
- ▶ Kronecker-Capelli Theorem
- ▶ Problems

Systems of Linear Equations

1 Definition

A general system of m linear equations with n unknowns can be written as

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n = b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n = b_2 \\ \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n = b_m \end{cases}$$

where $x_1, x_2, \dots, x_n$ are the unknowns, $a_{i,j} \in \mathbb{R}$ are the coefficients; $b_i \in \mathbb{R}$ are the constant terms.

A solution of a linear system is a tuple $(s_1, s_2, \dots, s_n)$ of numbers that makes each equation a true statement.

The **solution set** of the system is the set of all the solutions.

System of homogeneous linear equations

1 Definition

A system of homogeneous linear equations

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n &= 0 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n &= 0 \\ \vdots &\vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n &= 0 \end{cases}$$

always has a solution $u = (0, 0, \dots, 0)$. It is called a **trivial solution** (nghiệm tầm thường).

Augmented matrix (Ma trận mở rộng)

1 Definition

The **augmented matrix** $\tilde{A}$ of the general linear system is

$$\tilde{A} = [A \mid B] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right]$$

Note: A called the coefficient matrix, B is called the constant matrix.

Example 1. Find augmented matrix of the systems of linear equations

$$\begin{cases} x_1 - x_2 + 2x_3 + 4x_4 = 4 \\ 2x_1 + x_2 + 4x_3 = -3 \\ 2x_2 - 7x_3 = 5 \end{cases}$$

Example 2. Give the augmented

matrix $\bar{A} = \left(\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right)$

Find the systems of linear equations and it's solution.



Table of Contents

2 Solutions of Systems of linear equations

- ▶ Definition
- ▶ Solutions of Systems of linear equations
- ▶ Solving Systems of linear equations
- ▶ Kronecker-Capelli Theorem
- ▶ Problems

Solutions of Systems of linear equations

2 Solutions of Systems of linear equations

A system of equations is called **inconsistent** (không tương thích) if it has **no solutions**. It is called **consistent** (tương thích) **otherwise**.

Theorem

Any system of linear equations has one of the following exclusive conclusions.

1. No solution
2. Unique solution
3. Infinitely many solutions

Gaussian Elimination:

1. Perform elementary row operations to put the **augmented matrix** into the **row-echelon form**.
2. Solve the equation of the k th row for x_k , then substitute back into the equation of the $k - 1$ row to obtain a solution for x_{k-1} , etc.

Note: If you see the row $\left(0 \ 0 \ 0 \ 0 \ \dots \ 0 \ 0 \mid \neq 0 \right)$ then the systems of linear equations have **no solutions**.



Table of Contents

3 Solving Systems of linear equations

- ▶ Definition
- ▶ Solutions of Systems of linear equations
- ▶ Solving Systems of linear equations
- ▶ Kronecker-Capelli Theorem
- ▶ Problems



Example

3 Solving Systems of linear equations

Solve the linear equations

Example 1.

$$\begin{cases} 2x_1 + 3x_2 + 4x_3 = 3200 \\ x_1 + 2x_2 + x_3 = 1700 \\ x_1 + x_2 + 2x_3 = 1300 \end{cases}$$

Example 2.

$$\begin{cases} x + y + 2z = 1 \\ 2x + y + 3z = 2 \\ 3x + 2y + 5z = 3 \end{cases}$$

Example 3.

$$\begin{cases} 5x_1 - 2x_2 + 5x_3 - 3x_4 = 3 \\ 4x_1 + x_2 + 3x_3 - 2x_4 = 1 \\ 2x_1 + 7x_2 - x_3 = -1 \end{cases}$$



Table of Contents

4 Kronecker-Capelli Theorem

- ▶ Definition
- ▶ Solutions of Systems of linear equations
- ▶ Solving Systems of linear equations
- ▶ Kronecker-Capelli Theorem
- ▶ Problems

Kronecker-Capelli Theorem

4 Kronecker-Capelli Theorem

Theorem

If $\tilde{A} = [A \mid B]$ is the augmented matrix of the system with n unknowns, then $\text{rank}(\tilde{A}) = \text{rank}(A)$ or $\text{rank}(\tilde{A}) = \text{rank}(A) + 1$. Furthermore,

1. If $\text{rank}(\tilde{A}) = \text{rank}(A) + 1$ then the system has no solution.
2. If $\text{rank}(\tilde{A}) = \text{rank}(A) = n$ then the system has a unique solution.
3. If $\text{rank}(\tilde{A}) = \text{rank}(A) < n$ then the system has infinitely many solutions with $n - \text{rank}(A)$ free parameters.

Example. Solve and argue the system of linear equations by parameter m

$$\begin{cases} 3x_1 & + & 5x_2 & + & 3x_3 & - & 4x_4 & = & 1 \\ 2x_1 & + & 3x_2 & + & x_3 & + & x_4 & = & 0 \\ 5x_1 & + & 9x_2 & + & 6x_3 & - & 15x_4 & = & 2 \\ 13x_1 & + & 22x_2 & + & 13x_3 & - & 22x_4 & = & 2m \end{cases}$$



Table of Contents

5 Problems

- ▶ Definition
- ▶ Solutions of Systems of linear equations
- ▶ Solving Systems of linear equations
- ▶ Kronecker-Capelli Theorem
- ▶ Problems

Problems

5 Problems

1. Solve the following systems of linear equations

$$\text{a. } \begin{cases} x + y + 2z = 1 \\ 2x + 3y + z = -2 \\ 5x + 4y + 2z = 4 \end{cases}$$

$$\text{b. } \begin{cases} x_1 + 2x_2 - x_3 + x_4 = 0 \\ 2x_1 + 3x_2 - 2x_3 + 3x_4 = 0 \\ x_1 + x_2 - 3x_3 + x_4 = 0 \end{cases}$$

2. Find all values of k for which the system has nontrivial solutions.

$$\text{a. } \begin{cases} x - y + 2z = 0 \\ -x + y - z = 0 \\ x + ky + z = 0 \end{cases}$$

$$\text{b. } \begin{cases} x - 2y + z = 0 \\ x + ky - 3z = 0 \\ x - 6y + 5z = 0 \end{cases}$$

3. Determine the values of m such that the system has exactly one solution

$$\text{a. } \begin{cases} x - y + 2z = m \\ -x + y - z = 0 \\ -x + my - z = 1 - m \end{cases}$$

$$\text{b. } \begin{cases} mx + y + z = 1 \\ x + my + z = m \\ x + y + mz = m^2 \end{cases}$$

Problems

5 Problems

4. Determine the values of m such that the system of linear equation is inconsistent.

$$\text{a. } \begin{cases} x - y + 2z = m \\ -x + y - z = 0 \\ x - y + 3z = 1 - m \end{cases}$$

$$\text{b. } \begin{cases} x - 2y + 2z = m \\ x + my - z = 0 \\ 2x + y + mz = 2 - m \end{cases}$$

5. Solve the system of linear equation corresponding to the given augmented matrix

$$\tilde{A} = \begin{bmatrix} 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

6. Find a, b and c so that the system $\begin{cases} x + ay + cz = 0 \\ bx + cy - 3z = 1 \\ ax + 2y + bz = 5 \end{cases}$ has the solution $(3, -1, 2)$.

7. Find all values of k so that the system $\begin{cases} x + y - z = 2 \\ 2y + z = 3 \\ 4y - 2z = k \end{cases}$ has no solution.



Q&A

Thank you for listening!